

Beckett Frey

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EDUCATION

UW Milwaukee

B.S. Computer Science

Milwaukee, WI, USA

January 2023 - May 2025

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, Bash, SQL, R

Tools and Frameworks: Django, React, PyQt6, LangGraph, Node.js

Cloud and Infra: GitHub Actions, Docker, Modal, AWS

Speech and Audio ML: Forced Alignment, Whisper, Pyannote

Computer Vision: Object Detection (YOLO), Synthetic Data, mAP Evaluation

WORK EXPERIENCE

Software Engineer (Volunteer)

Environmental Data & Governance Initiative, Remote, USA

January 2026 - Present

- Fix small bugs and ship features which address analyst feedback for the web-monitoring project.
- Mitigate security concerns by switching to docker hardened images for release and dev phases.
- Modernize and improve continuous integration with ruff and git hooks, gating low quality code submissions from time consuming review.
- Collaborate with a team of volunteers across the country for environmental justice.

Research Intern

Waisman Center, WI, USA

September 2025 - Present

- Build continuous integration and deployment pipelines with github actions and docker across multiple code repositories to reduce problematic deployments and protect researchers time.
- Develop Python and Bash scripts to automate repetitive parts of the data collection pipeline.
- Collaborate across labs to support the ongoing development of legacy django Infrastructure, focusing on modernization and knowledge continuity.
- Code and market a cross-platform desktop application for phonetic research, helping onboard less technical researchers and automating metadata tracking for reproducibility.
- Provide technical support to researchers exploring machine learning for automating pronunciation assessment in children with speech sound disorders.

Junior ML Engineer

Hyperchess.ai, Remote, USA

December 2024 - September 2025

- Engineered and developed a synthetic data pipeline to train 25 YOLO models for piece detection, achieved 99% mAP@0.5 with a mixture of experts.
- Deployed inference workloads with Modal serverless GPU, drastically improving the speed of iteration.

PROJECTS

- **Phonetic Research Application**, Cross-platform desktop app that unifies AI/ML tools (MFA, Wav2TextGrid, Faster-Whisper, PyPLLRComputer) behind one interface, letting Speech Language Pathology researchers build and run phonetic analysis pipelines in one place. [Try it!](#)
- **Data Security Tooling**, Published PyPI tool that scans git repositories for PHI and other identifier patterns. Used to safely open-source several lab repos and caught one real PHI exposure in legacy code. [Try it!](#)
- **AI Document Generation**, Divides document generation into per-placeholder tasks queued by batch. Supports context injection and an enforced length range per task. [GitHub](#)
- **Multiplayer Chess API**, Chess backend in TypeScript/Node.js using WebSockets to sync game state and lobbies across clients. [GitHub](#)

EXTRACURRICULAR ACTIVITIES

- **Open Source Contributor**, Github - Ongoing
- **Runner**, Madison Marathon - November 2024
- **Bartender**, Ralley on the Runway - September 2024
- **Triathlete**, IRONMAN Wisconsin - September 2023